PROJECT SETTINGS

Project Name	:	G+2 RB BEAM DESIGN @SMR
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Beam-1-13.625 m.R1.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 00:54:36

General and Reinforcement Settings

Material Properties

Concrete Grade	:	M25	N/sqmm
Steel Grade (Main)	:	Fe550	N/sqmm
Steel Grade (Shear)	:	Fe550	N/sqmm
Top / Bottom Clear Cover	:	30	mm
Side Clear Cover	:	30	mm
Max Aggregate 2 Size	:	25	mm
Analysis Materials used	:	No	

Design Settings

Ignore Torsion	:	Yes	
Value Less Than	:	0.1	kNm
Flange Beam Considered	:	None	

Rebar Selected

Rebar Used	:	8,10,12,16,20,25,32,40

Main Rebar

Minimum	:	12
Maximum	:	20

Shear diameter

Minimum	:	8
Maximum	:	12

SFR Rebar

Minimum	:	8
Maximum	:	16

Stirrup Spacing

Minimum	:	75	mm
Maximum	:	300	mm

Spacing RoundOff	:	5	mm

Percentage Reinforcement

Minimum	:	0.2	%
Maximum	:	4	%
Nominal Reinforcement	:	0.2	%

Detail Settings

Top Detailing Style	:	Best Fit
Bottom Detailing Style	:	Best Fit

Crack Width Settings:

Crack Width IS 456 : 2000 + IS 13920 : 2016 Check	:	No

Curtailment Settings

Top Reinforcement Curtailment Location

End Support	:	L/4	
Clear Cover	:	30	mm
Continuous Support	:	L/4	

Bottom Reinforcement Curtailment Location

Mid Span	:	L/6
End Support	:	L/4
Continuous Support	:	L/4

Ductile Beam

Factor of Depth	:	Dx2

Non Ductile Beam

Factor of Length	:	L/3
Bending And Axial Force Design Settings	:	Not Consider

Beam Bar Spacing

Skip maximum spacing criteria between rebar for zero bending moment : No

Beam Width	Rebar Nos<=16	Spacing	Rebar Nos>=20	Spacing
230	3	61	2	130